Food Hub

Python Foundations

Date-Feb08-2026

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Executive Summary

Conclusion

- **Delivery time** or **preparation time**, **cost of order** or correlation between them are not influenced factors for customer rating
- **Ratings:** Not given" ratings form a large portion, indicating many customers skip rating.
- **Cost:** Orders above $20 generate higher revenue
- **Orders:** Generally pricing between $10–$20 covers most orders.
- **Cuisine** -Cuisines with high order costs yield high revenue, some cuisines show lower stable costs makes them budget-friendly and consistent. Preparation time varies by cuisine
- **Customers:** Customers in this area tend to place moderately priced orders. Customers usually spend around $10–$15 per order neither cheap nor expensive
- **Weekdays**: weekday delivery is slower than weekend deliveries

Recommendation

- Food Hub should support popular mid-range cuisines like American, Italian, Mexican that have consistent food quality and balanced pricing which results in good revenue
- Food hub should encourage cuisine like Thai and southern to have standard methods in food preparation which improve preparation time and customer satisfaction.
- Food hub should target premium cuisines like Mediterranean, Japanese which has high value customers who orders above $20 to generate more revenue
- Food Hub can provide promotional offers to encourage the customers to provide ratings.
- Food Hub needs to improve weekday delivery operations as it is slower than at weekends

Business Problem Overview and Solution Approach

Food Hub wants to understand what drives customer satisfaction and revenue, and to determine how to improve overall business efficiency and profitability.

Problem Definition

Although the company collects large amounts of data on orders, delivery times, preparation times, cuisine types, and customer ratings, it is unclear which factors influence customer experience and business performance

Unclear drivers of customer satisfaction

It is unknown whether delivery time, preparation time, or cost affects ratings.

Uneven restaurant performance

Some restaurants generate high revenue while others perform poorly. Food hub needs insight to support struggling restaurants

Operational ineffectiveness

To identify where delays occur and how to optimize operations.

Revenue Optimization

Which cuisine bring higher revenue
Which customers segments spend more
How to increase average order value

Solution Approach: Practice Data Analytics

1. **Understanding the data**

Explore number of records, types of variables, missing values

2. Data cleaning and preparation

Handling missing ratings, convert data types, Create new variables and columns

3. Exploratory Data Analysis

Use visualization to calculate summary statistics

Univariate Analysis – How each variable distributed independently

Bivariate Analysis – Two variables are dependent on and impacted one another

Correlation Analysis - Heatmap to check relationships between relevant factors

4. Business Metrics

Total Revenue, Percentages and average values to improve business performance

Data Overview

To analyze the data to get a fair idea about the demand of different restaurants which will help them in enhancing their customer experience, below is the data overview

- **order_id:** Unique ID of the order
- **customer_id:** ID of the customer who ordered the food
- **restaurant_name:** Name of the restaurant
- **cuisine_type:** Cuisine ordered by the customer
- **cost_of_the_order:** Cost of the order
- **day_of_the_week**: Indicates whether the order is placed on a weekday or weekend (The weekday is from Monday to Friday and the weekend is Saturday and Sunday)
- **rating:** Rating given by the customer out of 5
- **food_preparation_time:** Time (in minutes) taken by the restaurant to prepare the food. This is calculated by taking the difference between the timestamps of the restaurant's order confirmation and the delivery person's pick-up confirmation.
- **delivery_time:** Time (in minutes) taken by the delivery person to deliver the food package. This is calculated by taking the difference between the timestamps of the delivery person's pick-up confirmation and drop-off information
- **dataset name:** foodhub_order.csv

On analyzing the provided datasets

Q1. There are 1898 rows (records) and 9 columns (variables)

Q2. The data types of different columns in the dataset are float64(1), int64(4), object(4)

Columns	Data Types
order_id	int64
customer_id	int64
restaurant_name	String
cuisine_type	string
cost_of_the_order	float64
day_of_the_week	string
rating	string
food_preparation_time	int64
delivery_time	int64

Insight:

- order_id, customer_id, cost_of_the order, food prepration_time, delivery_time are columns holding numerical integers and float values
- restaurant_name, cuisine_type, day_of_the_week, rating are categorical String values

There are no missing values. Although 'Not given' values are present but they are not considered as NaN or null values

The statistical summary of numerical data is

Variable	count	mean	std	min	25%	50%	75%	max
cost_of_the_order	1898	1.64	7.5	4.47	12.08	14.14	2.29	35.41
food_preparation_tima	1898	2.7	4.6	20.00	23.00	27.00	3.1	35.00
delivery_time	1898	2.4	5	15.00	20.00	25.00	2.8	33.00

Insight:

- preparation ranges minimum of 20 minutes to a maximum of 35 minutes with average preparation time of 27minutes
- delivery time ranges minimum of 15 minutes to maximum of 33 min with average delivery time of 25 minutes

Ratings:

Rating	count
Not given	736
5	588
4	386
3	188

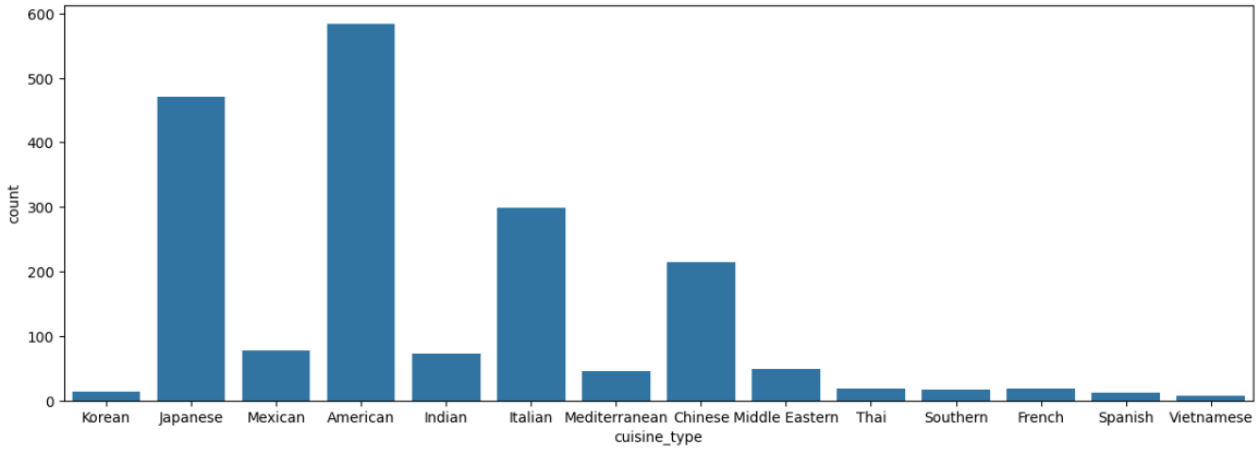
Insight: 736 orders are not rated

Univariate Analysis

Univariate analysis has been performed on variables, provided observations explored their distributions using charts like histograms, boxplot, count plots, bar chart

- There are 1898 unique order_id.
- There are about 1200 rows i.e. 1200 unique customer_id. The remaining 698 records may be duplicated, which means customers re visited 698 times.
- There are about 178 unique restaurants and 14 unique Cuisine types located in NYC area.

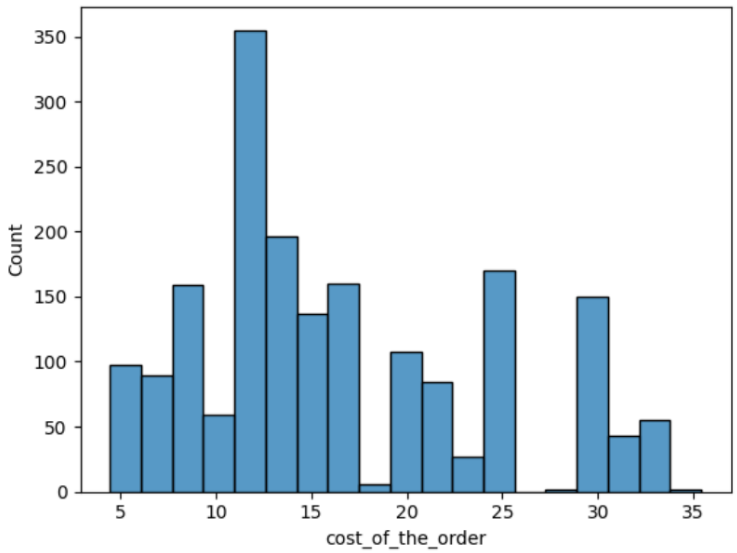
<u>Cuisine type</u>

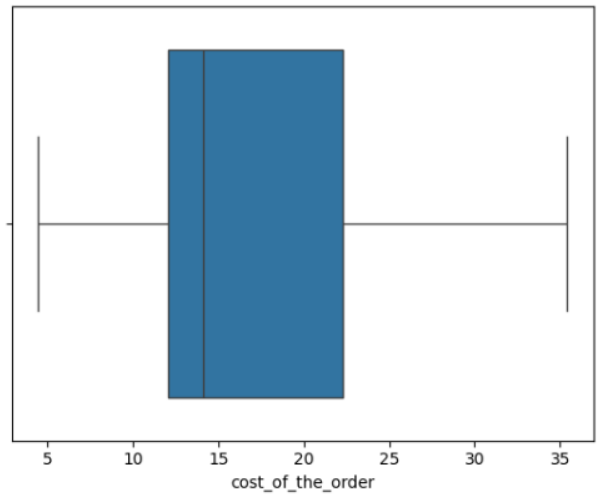


Insight:

- American restaurants received the highest number of orders in the dataset followed by Japanese and Italian Chinese, but American is far ahead.
- Korean Spanish, Vietnamese cuisines have lowest counts
- Mexican, Indian, Mediterranean, Thai southern is stable and balanced supply-demand

Cost of the Order

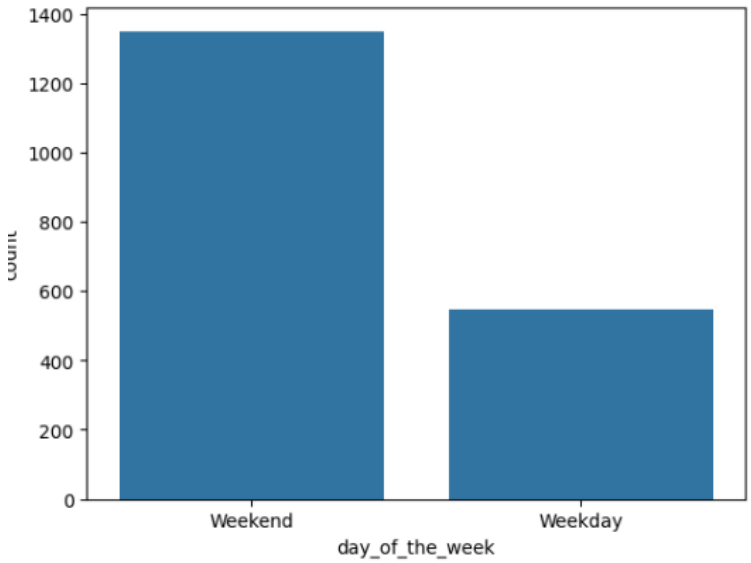




Insight:

- Most Orders fall into ($10-$18) dollars
- The distribution is slightly right-skewed, which means fewer orders cost high at the range of ($25-$30) dollars
- Very low-cost order (<$5) is rare. Only small number of orders falls in lowest price range and contributes no revenue
- Median order cost is around $15
- The overall cost range is roughly $5 to $35

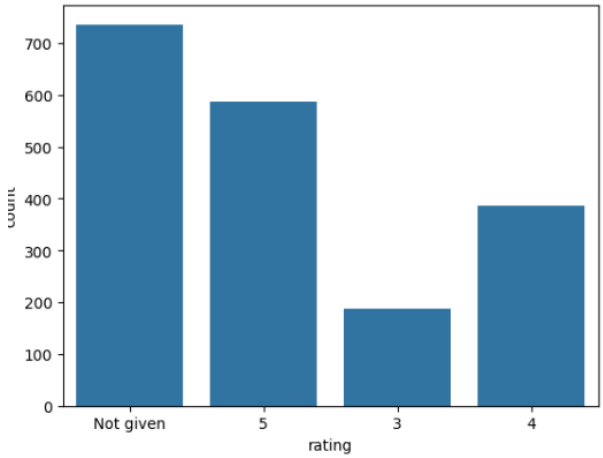
Day of the Week



Insight:

- Weekend orders are approximately 1350. Weekday orders are approximately 550
- Weekend orders are more than double weekday orders

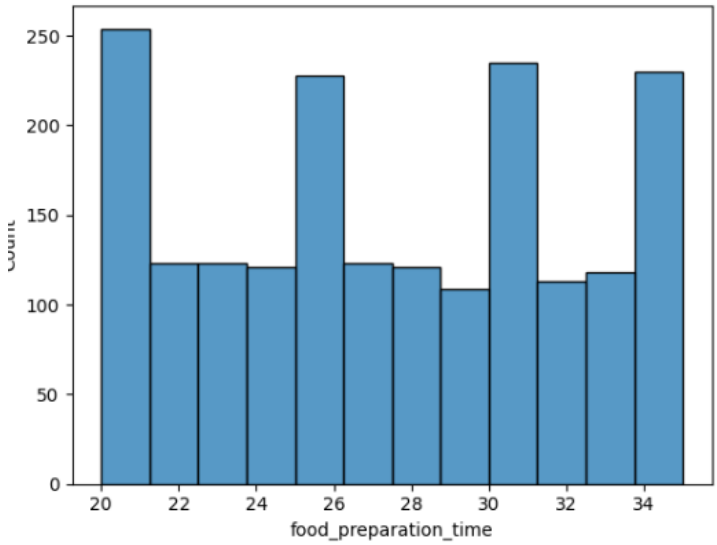
Rating



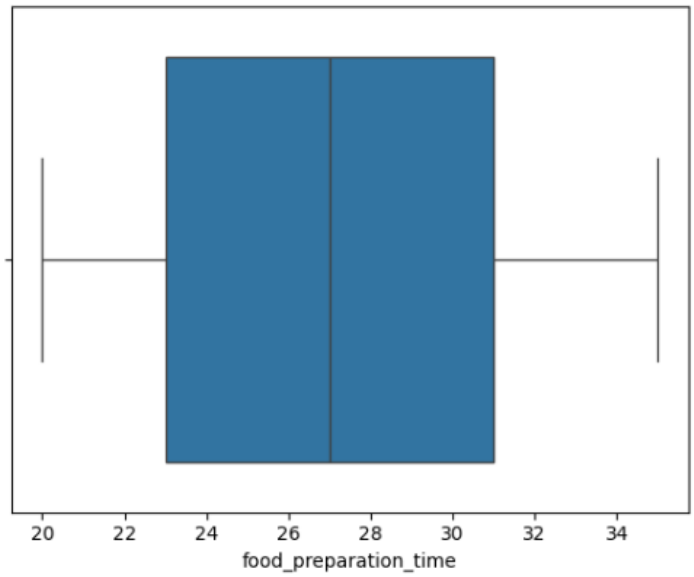
Insight

- A significant portion of customers did not provide a rating after ordering. almost 720 orders marked as Not given are not rated
- The next highly rated is '5' with around 500 ratings

Food Preparation Time

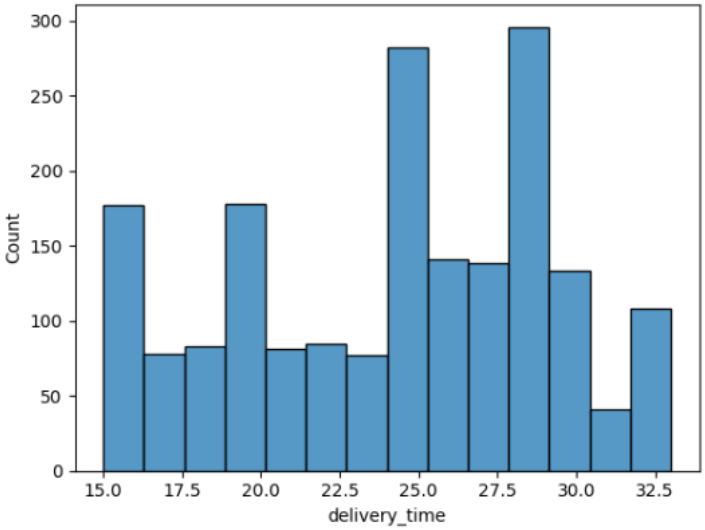


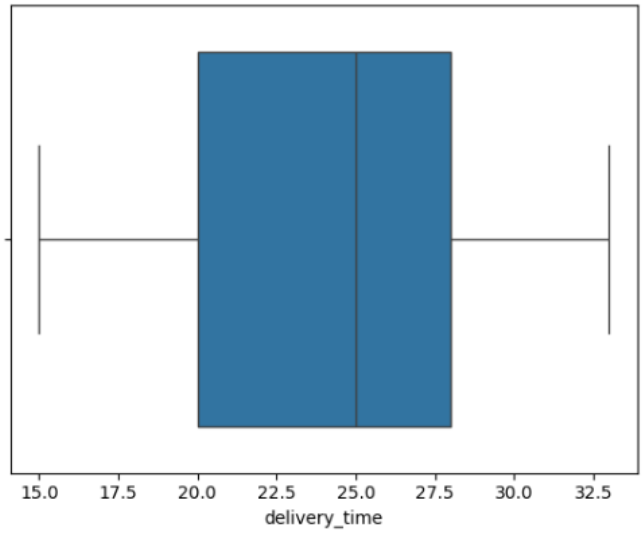
Preparation time mostly falls between 20 and 35 minutes
The distribution is fairly balanced



- Everything stays within a tight range (20–35 minutes). There is not extremely quick or extreme delay in preparation time
- Lower quartile (Q1) ≈ 23 minutes
- Minimum ≈ 20 minutes, maximum ≈ 35 minutes
- Upper quartile (Q3) ≈ 32 minutes
- Median prep time ≈ 27 minutes

Delivery time





Insight:

- Delivery time cluster around 25-28 minutes
- Very fast deliveries (15-20 minutes) and very slow deliveries (30+ minutes) are less common.
- This indicates a stable and reliable delivery system with consistent customer waiting times.

Q7-The top 5 restaurants in terms of orders received are

Restaurant_name	count
Shake Shack	219
The Meatball Shop	132
Blue Ribbon Sushi	119
Blue Ribbon Fried Chicken	96
Parm	68

Insight: The highest number of orders (219) is received by 'Shake Shack' and then followed by sudden drop in the count of orders by the meatball shop (132 orders), Blue Ribbon Sushi (119 orders), Blue Ribbon Fried Chicken (96) and Parm (68)

Q-8. Among the cuisines Korean, Japanese, American, Italian, Mexican, Mediterranean, Chinese, Indian, Thai, Southern, French, Spanish, Middle Eastern, Vietnamese

Cuisine type	count
American	415
Japanese	335
Italina	207
Chinese	163
Mexican	53
Indian	49
Middle Eastern	32
Mediterranean	32
Thai	15
French	13
Korean	11
Southern	11
Spanish	11
Vietnamese	4

Insight: The most popular cuisine is American with 415 number of orders

Q9. The total number of orders above $20 is **555**

The percentage of orders cost more than 20 dollars is **29.24%**

Q10. The mean delivery time is **24.16 minutes**

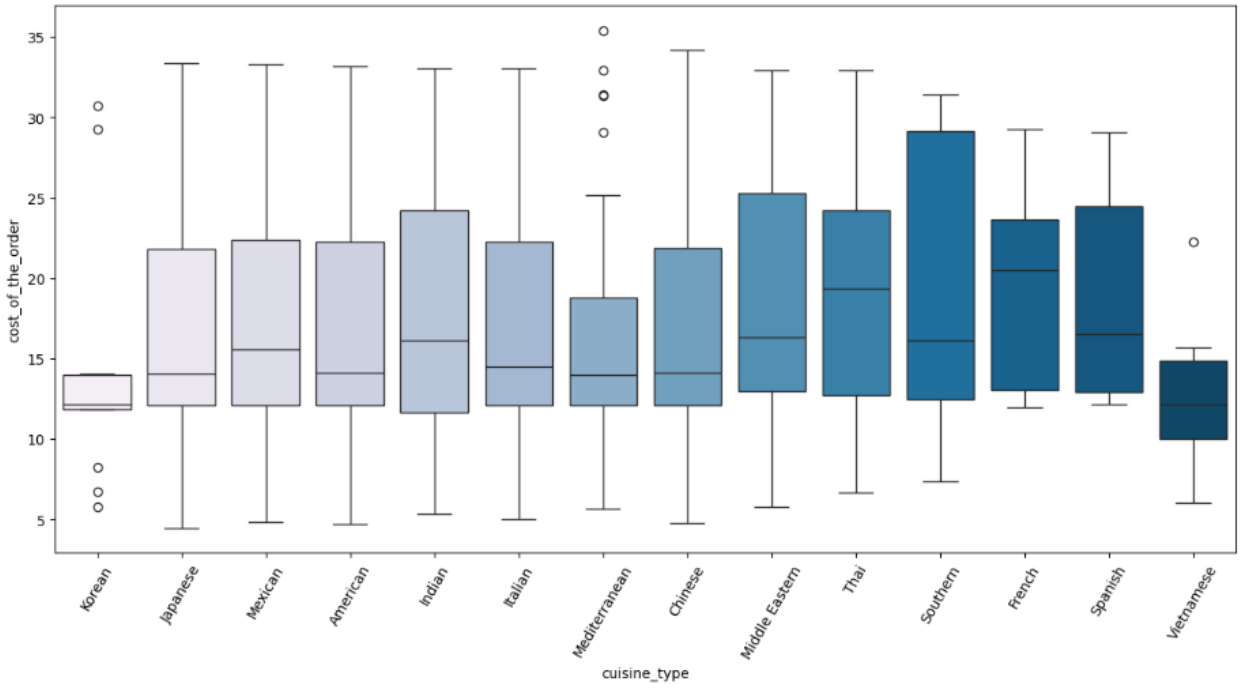
Q11. If company decided to give 20% discount voucher to top 5 most frequent customers, then it should be given to below customers id

- 52832 who ordered 13 times
- 47440 who ordered 10 times
- 83287 who ordered 9 times
- 250494 who ordered 8 times
- 259341 who ordered 7 times

Multivariate Analysis

Multivariate analysis is performed to explore relationships between multiple numerical variables as well as relations between numerical and categorical variables

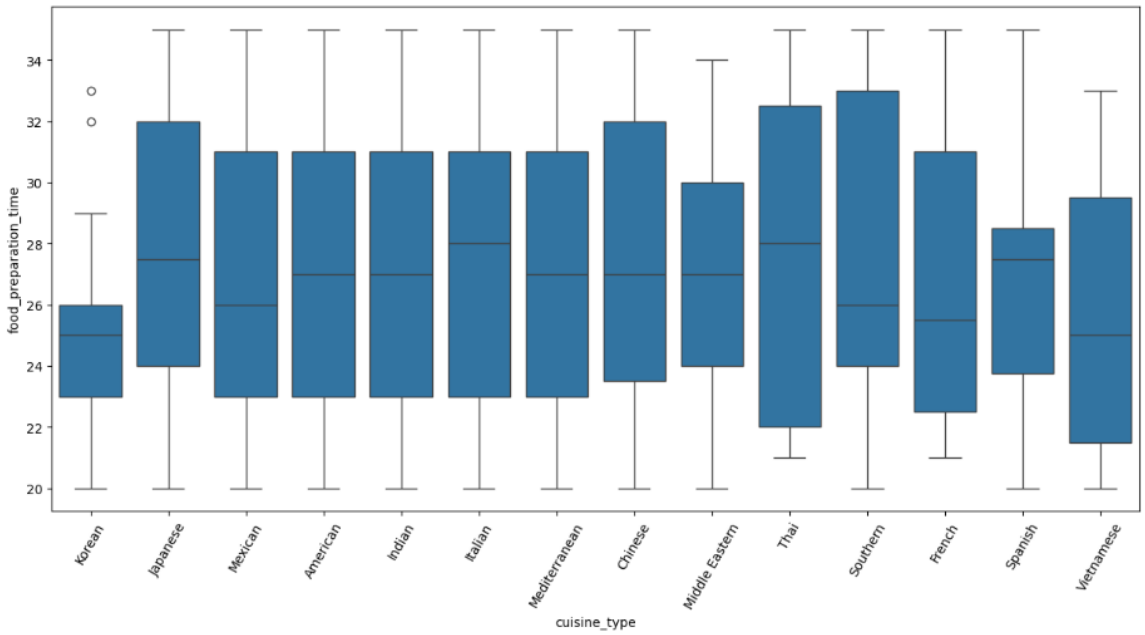
<u>Cuisine vs Cost of the order</u>



Insight:

- Premium cuisines: Italian, Japanese, Mediterranean shows approximate range from 20 to 18 show higher spending
- Mid to upper pricing: American and Italian ranges (5 -30) $, has wide range of spending with consistent demand
- Mid-priced: Mexican, Indian, Chinese shows $ 11-13 app. budget- friendly
- Korean, Spanish and Vietnamese are the most affordable cuisines

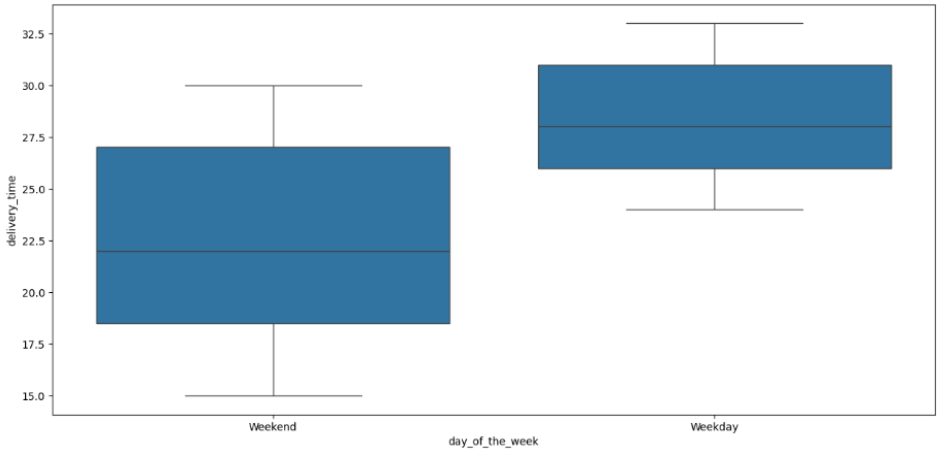
Cuisine Vs Food Preparation time



Observation:

- Korean with 23-26 min and it has consistent prep times with narrow IQR.
  So Korean dishes are fastest
- Thai and southern cuisines show wide variability because some dishes take longer time to prepare
- Japanese cuisine has moderate preparation time with occasional long outliers
- American, Italian, Mediterranean, Chinese, and Indian cuisines have median around 25minutes
- Most of the restaurants are generally taken to 23-32min to prepare

Day of the week vs Delivery time



- Deliveries are faster on weekends while weekdays experience delays
- weekend median is 18-27 minutes
- weekday median is 26-30 minutes
- That's roughly a 6–7-minute increase on weekdays

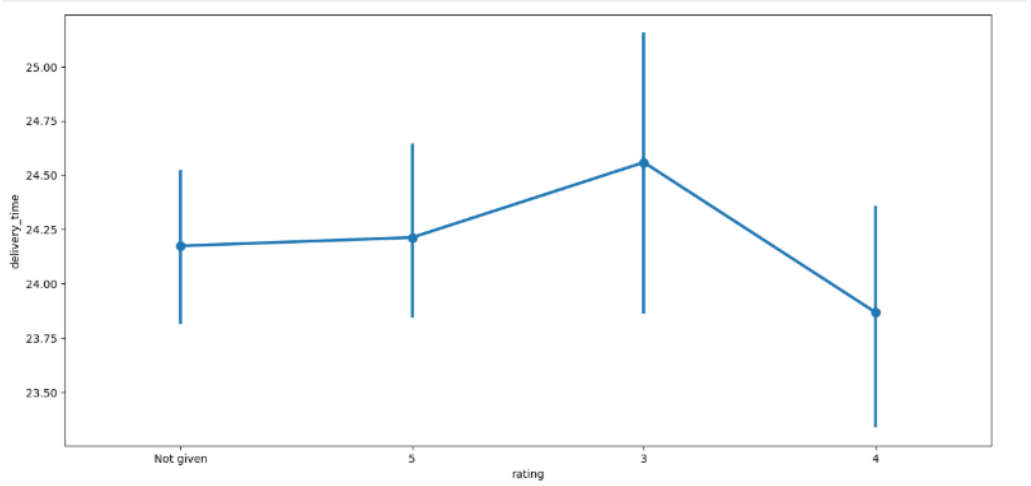
Restaurant vs cost of the order

Restaurant name	Cost of the order
Shake shack	3579.53
The meatball shop	2145.21
Blue Ribbon Sushi	1903.95
Blue Ribbon fried chicken	1662.29
Parm	1112.76
RedFarm Broadway	965.13
RedFarm Hudson	921.21
TAO	834.50
Han Dynasty	755.29
Blue Ribbon Sushi Bar & grill	666.62
Rubirosa	660.45
Sushi of Gari 46	640.87
Nobu Next door	623.67
Five Guys Burgers and Fries	506.47

Insight:

- Shake shack is the dominant revenue generated by large margin.
- Revenue declines sharply after the top four restaurants.
- American, Japanese, Italian, followed by mediterranean are high revenue restaurants.

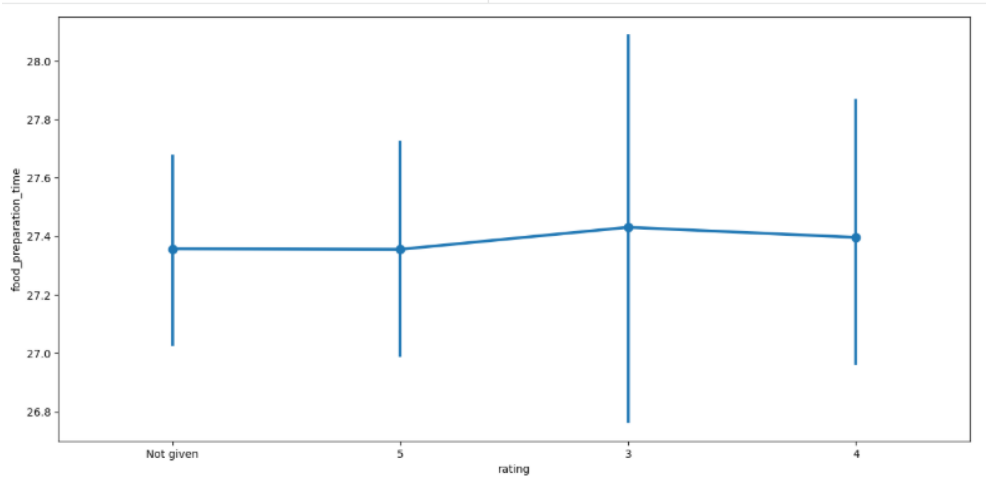
Rating vs Delivery time



Observation:

- Delivery times are significantly longer on weekdays- higher median and whisker extend to higher values
- Weekend delivery times are shorter and more efficient- Lower median and narrow interquartile range

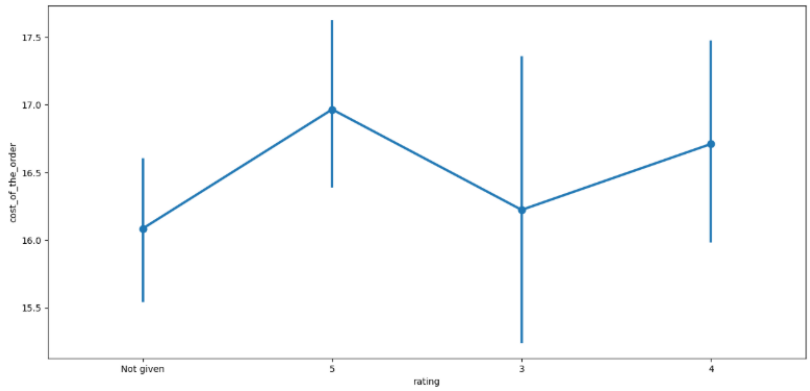
Rating vs Food Preparation



Observation:

- Not given- is also given to faster preparation time
- Food preparation time is almost the same across all rating categories between 27-28 minutes
- Rating 3 has slightly longer prep time
- Rating 5 is not based on preparation time

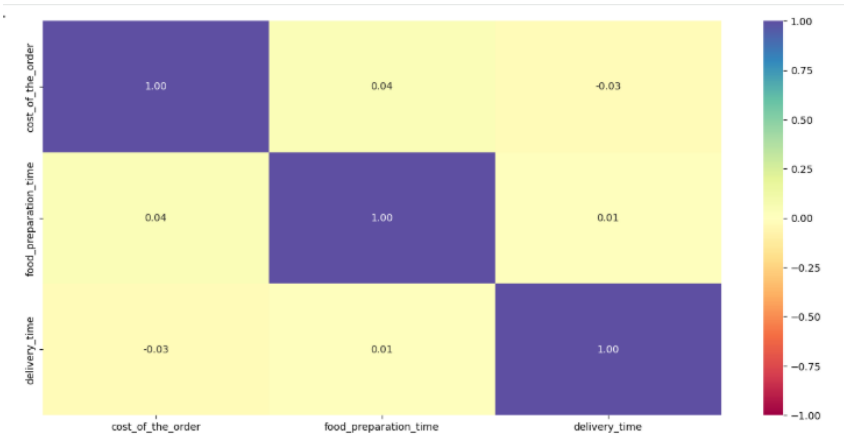
Rating Vs Cost of the order



Observation:

- All points lie between ~15.5 and ~17.5 dollars, showing very small variation
- Rating 5 has a slightly higher average cost of around $16.8
- Rating 3 is given to order with cost of ≈ $16.5
- Rating 4 is given to order with cost of ≈ $16.3
- Customers who don't rate (Not rated) tend to place slightly cheaper orders.

Correlation among variables



Observation:

- The corr. Relation between cost – preparation time is 0.01
- The corr. Relation between preparation time- delivery time 0.01
- cost-delivery time is negative 0.03
- correlation between them is negligible
- the three operational variables behave independently

Q13. The company wants to provide a promotional offer, for which the restaurants must have a rating of more than 50 and the average rating should be greater than 4.

Restaurant name	Rating counts
Shake Shack	133
The Meatball Shop	84
Blue Ribbon Sushi	73
Blue Ribbon Fried Chicken	64
Red Fram Broadway	41

Observation:

Totally 1162 orders are rated. Now among which shortlist the restaurant names which has ratings count more than 50. It is shortlisted to four restaurants by filtering out by choosing rating as greater than 4.

Restaurant name	Rating
The Meatball Shop	4.5
Blue Ribbon Fried Chicken	4.32
Shake Shack	4.27
Blue Ribbon Sushi	4.22

Observation:

It is further narrowed down to 4 restaurants as

The Meatball Shop, Blue Ribbon Fried Chicken, shake shack, Blue Ribbon Sushi. These restaurants are eligible to get promotional offers

Q14. The company charges the restaurant 25% on the orders having cost greater than 20 dollars and 15% on the orders having cost greater than 5 dollars

Order_id	Cost_of_the_order	Revenue
1477147	30.75	7.68
1477685	12.08	1.81
1477070	12.23	1.83
1477334	29.20	7.3
1478249	11.59	1.74

Observation- The net revenue generated by the company is 6166 dollars

Q15. The company wants to analyze the total time required to deliver the food. What percentage of orders take more than 60 minutes to get delivered from the time the order is placed?

Observation: There are 200 orders and percentage of orders that take more than 60 minutes is 10.54%

Q16. The company wants to analyze the delivery time of the orders on weekdays and weekends. How does the mean delivery time vary during weekdays and weekends?

Observation:

- The mean delivery time on weekdays is around 28 minutes
- The mean delivery time on weekdays is around 22 minutes